Supplementary materials

Comparative efficacy and safety of systemic treatments for cutaneous leishmaniasis: a systematic review and network meta-analysis

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Appendix S1

Complete search strategies including as follows:

PubMed

((“systemic treatment*”[tiab] OR “systemic therapy”[tiab] OR “pentavalent antimon*”[tiab] OR therapeutics[tiab] OR “drug therapy”[tiab] OR “azole molecules”[tiab] OR “amphotericin B”[tiab] OR miltefosine[tiab] OR “pentamidine isethionate”[tiab] OR paromomycin[tiab] OR immunotherapy[tiab] OR immunomodulat*[tiab] OR GM-CSF[tiab] OR “granulocyte macrophage colony-stimulating factor”[tiab] OR “zinc therapy”[tiab] OR “sodium stibogluconate”[tiab] OR “meglumine antimoniate”[tiab] OR fluconazole[tiab] OR ketoconazole[tiab] OR itraconazole[tiab] OR cyclosporin[tiab] OR “dihydrocyclosporin A”[tiab] OR Imiquimod[tiab])
AND
(“Leishmaniasis, Cutaneous”[mesh] OR “cutaneous leishmaniasis”[tiab] OR “cutaneous leishmaniases”[tiab] OR “leishmaniases, cutaneous”[tiab] OR “leishmaniasis, old world”[tiab] OR “old world leishmaniasis”[tiab] OR “leishmaniasis, new world”[tiab] OR “new world leishmaniasis”[tiab])
AND
(“randomized controlled trial”[mesh] OR “controlled clinical trial”[mesh] OR randomly[tiab] OR randomized[tiab] OR trial[tiab] OR “cohort studies”[tiab]))

Scopus

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KEY (ketoconazole) OR TITLE-ABS-KEY (itraconazole) OR TITLE-ABS-KEY (cyclosporin) OR TITLE-ABS-KEY ("dihydrocyclosporin A") OR TITLE-ABS-KEY (imiquimod) AND TITLE-ABS-KEY ("Leishmaniasis, Cutaneous") OR TITLE-ABS-KEY ("cutaneous leishmaniasis") OR TITLE-ABS-KEY ("cutaneous leishmaniases") OR TITLE-ABS-KEY ("leishmaniases, cutaneous") OR TITLE-ABS-KEY ("leishmaniasis, old world") OR TITLE-ABS-KEY ("old world leishmaniasis") OR TITLE-ABS-KEY ("leishmaniasis, new world") OR TITLE-ABS-KEY ("new world leishmaniasis") AND TITLE-ABS-KEY ("randomized controlled trial") OR TITLE-ABS-KEY ("controlled clinical trial") OR TITLE-ABS-KEY (randomly) OR TITLE-ABS-KEY (randomized) OR TITLE-ABS-KEY (trial) OR TITLE-ABS-KEY ("cohort studies")) View less

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('systemic treatment*':ti,ab,kw OR 'systemic therapy':ti,ab,kw OR 'pentavalent antimon*':ti,ab,kw OR therapeutics:ti,ab,kw OR 'drug therapy':ti,ab,kw OR 'azole molecules':ti,ab,kw OR 'amphotericin b':ti,ab,kw OR miltefosine:ti,ab,kw OR 'pentamidine isethionate':ti,ab,kw OR paromomycin:ti,ab,kw OR immunotherapy:ti,ab,kw OR immunomodulat*:ti,ab,kw OR 'gm csf':ti,ab,kw OR 'granulocyte macrophage colony-stimulating factor':ti,ab,kw OR 'zinc therapy':ti,ab,kw OR 'sodium stibogluconate':ti,ab,kw OR 'meglumine antimoniate':ti,ab,kw OR fluconazole:ti,ab,kw OR ketoconazole:ti,ab,kw OR itraconazole:ti,ab,kw OR cyclosporin:ti,ab,kw OR 'dihydrocyclosporin a':ti,ab,kw OR imiquimod:ti,ab,kw) AND ('leishmaniasis, cutaneous':ti,ab,kw OR 'cutaneous leishmaniasis':ti,ab,kw OR 'cutaneous leishmaniases':ti,ab,kw OR 'leishmaniases, cutaneous':ti,ab,kw OR 'leishmaniasis, old world':ti,ab,kw OR 'old world leishmaniasis':ti,ab,kw OR 'leishmaniasis, new world':ti,ab,kw OR 'new world leishmaniasis':ti,ab,kw) AND ('randomized controlled trial':ti,ab,kw OR 'controlled clinical trial':ti,ab,kw OR randomly:ti,ab,kw OR randomized:ti,ab,kw OR trial:ti,ab,kw OR 'cohort studies':ti,ab,kw)

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AND
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AND
(“randomized controlled trial” OR “controlled clinical trial” OR randomly OR randomized OR trial OR “cohort studies”))

Supplementary figures

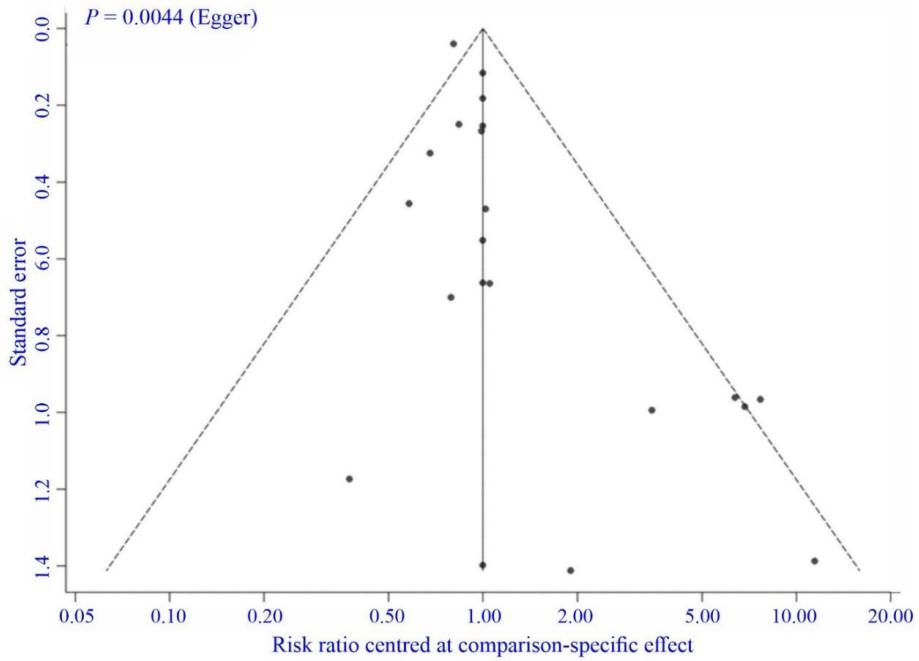


Fig. S1. Comparison-adjusted funnel plot assessing small-study effects in the network meta-analysis excluding high-risk studies.

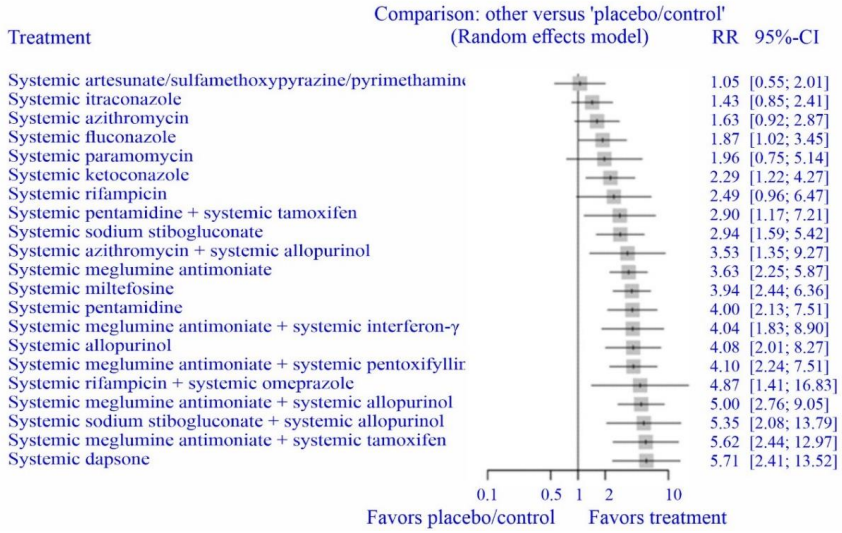


Fig. S2. Forest plot from a network meta-analysis comparing systemic treatments against placebo/control for cutaneous leishmaniasis. RR with 95% CI are shown. The horizontal line denotes the 95% CI, and the vertical line indicates no effect (RR = 1). The I^2 statistic for heterogeneity was 68.8%. RR, Relative risk; CI, confidence interval.

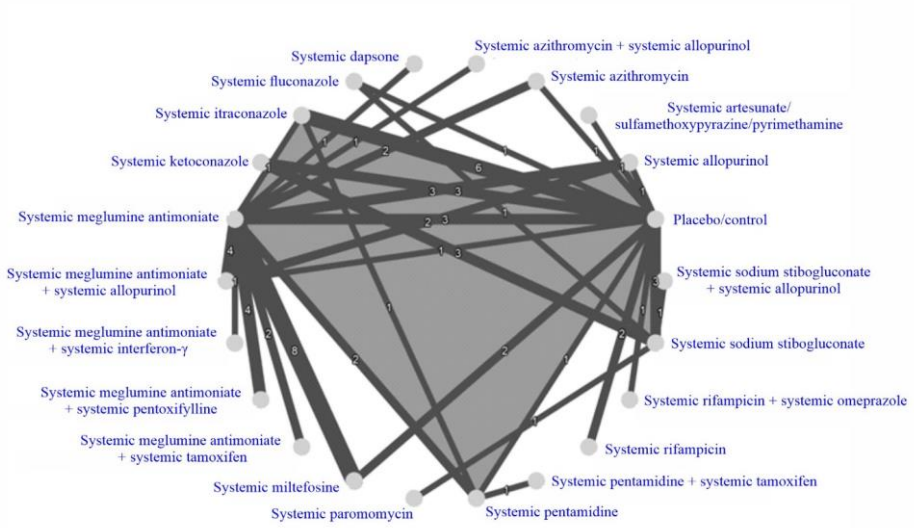


Fig. S3. Network plot of systemic treatments for cutaneous leishmaniasis. Node size represents the number of participants/intervention arms, and line thickness represents the number of direct comparisons between treatments.

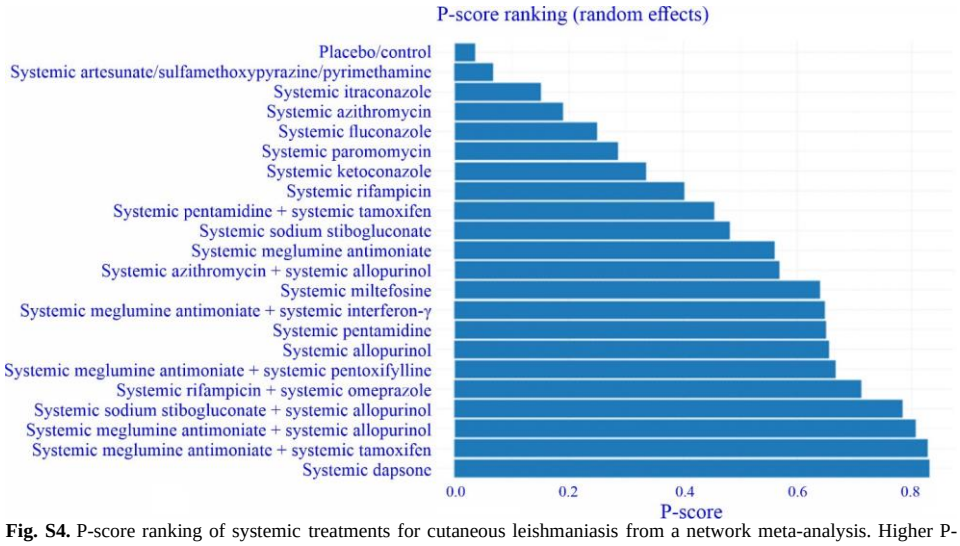


Fig. S4. P-score ranking of systemic treatments for cutaneous leishmaniasis from a network meta-analysis. Higher P-scores indicate greater confidence in a treatment's efficacy. Rankings are shown for the random-effects model. Systemic dapsone achieves the highest rank, followed by combination therapies such as meglumine antimoniate + tamoxifen, sodium stibogluconate + allopurinol, and meglumine antimoniate + allopurinol. Placebo/control consistently ranks lowest.

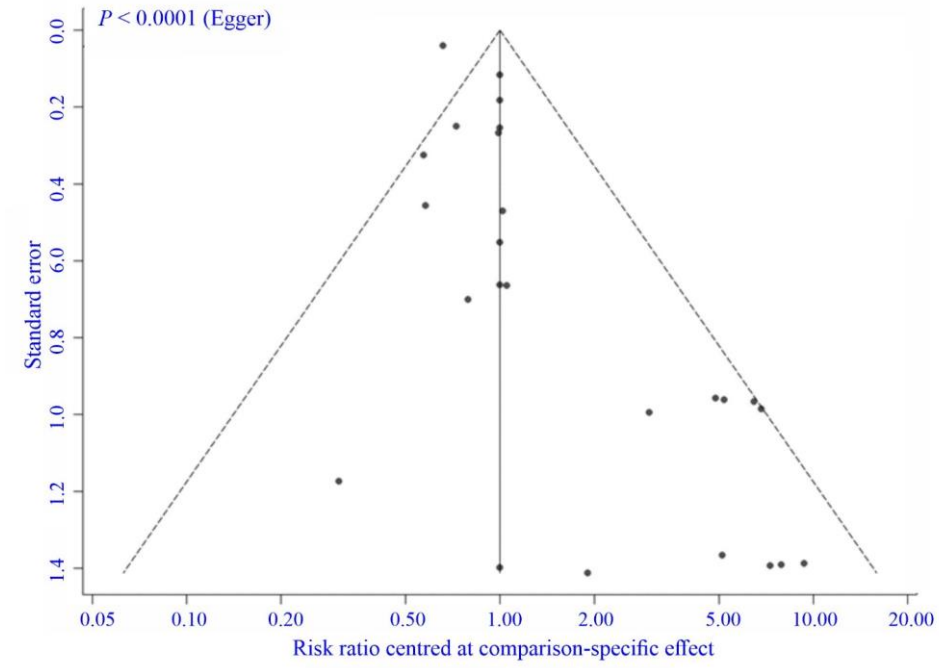


Fig. S5. Comparison-adjusted funnel plot assessing small-study effects in the overall network meta-analysis.

Supplementary tables

Table S1. Baseline characteristics of RCTs included in qualitative analysis.

Study	Age range or mean (year)	*Leishmania* Species	Intervention (number of participants)	Cure rate (% per group)	Follow-up (weeks)	AEs
Abtahi-Naeini *et al.*, 2021 (1)	25.9	*L. major*	Oral azithromycin (14, 20 lesions)	65%	12	No significant AEs
	25.1		Oral + topical liposomal azithromycin (13, 21 lesions)	85.7%		Mild pruritus
Adam *et al.*, 2009 (2)	30.6	Not specified	Oral artesunate + sulfamethoxypyrazine/pyrimethamine (20)	90%	10	No significant AEs
Ahmed *et al.*, 2022 (3)	31 (20-80)	Not specified	IM meglumine antimoniate (50)	42%	11	Four treatment discontinuations due to AE
			Oral dapsone (50)	66%		Five treatment discontinuations due to AE
Alfahaad, 2020 (4)	5-65	Not specified	IM + IL sodium stibogluconate (20 patients)	60%	20	Systemic AEs were absent
Al-Fouzan *et al.*, 1991 (5)	12-52	Not specified	Oral itraconazole (15)	73%	12	Nausea, headache, and elevated liver enzymes
Alkhawajah *et al.*, 1997 (6)	13-42	*L. major*	IM meglumine antimoniate (31, 68 lesions)	68%	4	No significant AEs
Almeida *et al.*, 1999 (7)	16-51	*L. braziliensis*	IV sodium stibogluconate (10)	100%	42	No significant AEs
			IV sodium stibogluconate + IL GM-CSF (10)	100%		Muscle aches and general malaise
Alrajhi *et al.*, 2002 (8)	31.2	*L. major*	Oral fluconazole (80)	79%	12	Mild AEs
Arana *et al.*, 1994 (9)	20.1	*L. braziliensis* and *L. mexicana*	IV meglumine antimoniate for 20 days (21)	90%	52	Mild AEs
	18.9		IV meglumine antimoniate for 10 days (20)	90%		Mild AEs
	19.6		IV meglumine antimoniate + interferon-γ for 10 days (22)	100%		Mild AEs such as mild malaise, headache and fever
Arevalo *et al.*, 2007 (10)	34.9	Not specified	IV meglumine antimoniate (7)	57%	12	Mild AEs
			IV meglumine antimoniate + imiquimod cream (7)	100%		Mild to moderate erythema, edema and sensations of burning
Armijos *et al.*, 2004 (11)	21.3	*Leishmania* from the *Viannia* subgenus	IM meglumine antimoniate (36)	91.7%	52	Systemic AEs such as fever, headache and weakness

Aronson *et al.*, 2010 (12)	18-57	*L. major*	IV sodium stibogluconate (26)	54%	52	Reversible abdominal pain/pancreatitis, arthralgias, myalgias, headache, fatigue, and elevated liver enzymes.
Bashir *et al.*, 2019 (13)	16-60	Not specified	Oral itraconazole + IL meglumine antimoniate (35)	100%	12	No significant AEs
Brito *et al.*, 2014 (14)	34 29	*L. braziliensis*	IV meglumine antimoniate (15) IV meglumine antimoniate + oral pentoxifylline (18)	47% 56%	12	Mild AEs, such as arthralgia, headache, and fever, were more common in the pentoxifylline group
Brito *et al.*, 2017 (15)	18-62	*L. braziliensis*	IV meglumine antimoniate (82) IV meglumine antimoniate + oral pentoxifylline (82)	43% 45%	24	The pentoxifylline group had 37.8% AEs compared to 23% for the other group
Castro *et al.*, 2022 (16)	18-62	*L. panamensis*, *L. braziliensis* and *L. guyanensis*	IM meglumine antimoniate (36) IM meglumine antimoniate + oral pentoxifylline (34)	75% 64.7%	26	Similar AEs, with fever, headache, elevated liver enzymes, and injection site responses being the most common
Celes *et al.*, 2022 (17)	19-42	*L. braziliensis*	IV meglumine antimoniate (9) IV meglumine antimoniate + topical bacterial cellulose bio-curatives (10)	55% 80%	12	No AEs were reported for either group
Chrusciak-Talhari *et al.*, 2011 (18)	4-62	*L. guyanensis*, *L. braziliensis* and *L. lainsoni*	Oral miltefosine (56) IV meglumine antimoniate (28)	71.4% 53.6%	24	48.3% reported vomiting, 8.6% nausea, 6.7% diarrhea, and 3.4% orchitis 33% reported arthralgias
D'Oliveira Jr. *et al.*, 1997 (19)	12-45	*L. braziliensis*	Oral allopurinol (18) IV meglumine antimoniate (16)	0% 50%	52	Four patients in the allopurinol group experienced worsened ulcers, edema, and pain. One patient developed mucosal disease
Dastgheib *et al.*, 2012 (20)	39.7 36.8	*L. major*	Oral azithromycin + allopurinol (36) IM meglumine antimoniate (35)	38.9% 40%	8	Minimal AEs. One patient developed gastrointestinal issues Minimal AEs. two patients experienced myalgia
Dogra *et al.*, 1990 (21)	14-56	*L. major* and *L. tropica*	Oral itraconazole (15)	66.6%	12	Three patients experienced mild headache
Dogra *et al.*, 1994 (22)	15-62	*L. major* and *L. tropica*	Oral itraconazole (20)	75%	24	Three patients experienced transient liver function test abnormalities
Dogra *et al.*, 1996 (23)	19-62	*L. major* and *L. tropica*	Oral itraconazole (10)	80%	12	One patient experienced elevated liver enzymes and

						another patient reported nausea
El-Sayed *et al.*, 2010 (24)	12-50	Not specified	IM + IL sodium stibogluconate (10, 15 lesions) Oral ketoconazole + IL sodium stibogluconate (10, 13 lesions)	93.3% 92.3%	24	No systemic AEs were observed
Esfandiarpour *et al.*, 2002 (25)	14	Not specified	Oral allopurinol (50) IM meglumine antimoniate (50) Oral allopurinol + IM meglumine antimoniate (50)	18% 24% 46%	4	Nausea, heartburn, and mild increases in liver enzymes only in allopurinol group
Farajzadeh *et al.*, 2015 (26)	20.74 16.3	*L. tropica*	IM meglumine antimoniate + cryotherapy (40) Oral terbinafine + cryotherapy (40)	52.5% 37.5%	12	No major AEs were observed in either group
Firooz *et al.*, 2006 (27)	27	*L. tropica*	IM meglumine antimoniate + imiquimod cream (59) IM meglumine antimoniate (60)	44.1% 40%	20	Three patients experienced pruritus and burning sensation No AEs were reported
Hanif *et al.*, 2016 (28)	39	Not specified	Oral chloroquine (38)	100%	16	No systemic AEs
Hepburn *et al.*, 1994 (29)	23.8 23.5	*L. mexicana* and *L. braziliensis*	IV paromomycin (17) IV sodium stibogluconate (17)	59% 88%	24	Well tolerated with mild AEs such as muscle aches and a papular rash in one patient More significant AEs, including muscle aches, nausea, and bone marrow suppression
Jaffar *et al.*, 2006 (30)	3-65	Not specified	Oral rifampicin (34)	61.8%	12	One patient had transient elevated liver enzymes
Khattak *et al.*, 2021 (31)	27.4	*L. tropica*	Oral allopurinol + IV meglumine antimoniate (56) IV meglumine antimoniate (56)	83.9% 73.2%	8	No major AEs were reported in either group
Kochar *et al.*, 2000 (32)	29.5	Not specified	Oral rifampicin (23)	73.9%	4	No major AEs
Kochar *et al.*, 2006 (33)	16-70	*L. tropica*	Oral rifampicin + omeprazole (23)	69.6%	6	No major AEs
Krolewiecki *et al.*, 2007 (34)	33 37	*L. braziliensis*	Oral azithromycin (22) IV or IM meglumine antimoniate (23)	45.5% 82.6%	52	Well-tolerated with mild gastrointestinal AEs 78% experienced musculoskeletal AEs
Lago *et al.*, 2024 (35)	18-50	*L. braziliensis*	IV meglumine antimoniate + topical rSm29 (31) IV meglumine antimoniate (30)	71% 43%	12	56% experienced joint and muscle pain, with some cases of mild itching and small papular lesions that self-healed 54% experienced joint and muscle pain

Layegh *et al.*, 2007 (36)	21.77 22.78	*L. tropica* and *L. major*	Oral azithromycin (20, 29 lesions) IM meglumine antimoniate (27, 58 lesions)	10.3% 34.5%	16	Two patients experienced gastrointestinal symptoms (nausea, vomiting), leading to discontinuation Three patients reported myalgia and erythema at the injection site
Lobo *et al.*, 2006 (37)	36	*L. braziliensis*	IV meglumine antimoniate (20)	10%	4	Mild secondary bacterial infections in one patient
López *et al.*, 2012 (38)	Not specified	*L. braziliensis* and *L. panamensis*	IM meglumine antimoniate (121)	85%	24	Myalgia, arthralgia, headache, and fever
López *et al.*, 2022 (39)	34.3	*L. braziliensis* and *L. panamensis*	Oral miltefosine + thermotherapy (62)	83.9%	24	Vomiting (31.8%) and liver enzyme elevation (28.8%)
Lyra *et al.*, 2023 (40)	≥ 13	Not specified	IV or IM meglumine antimoniate (59)	67.8%	24	80.6% clinical AEs, 73.1% laboratory AEs, and 25.4% electrocardiographic AEs. Ten participants were discontinued due to severe or persistent AEs
Machado *et al.*, 2010 (41)	22 23	*L. braziliensis*	Oral miltefosine (60) IV meglumine antimoniate (30)	75% 53.3%	24	Vomiting, nausea, and abdominal pain with 78.3% rate of AEs Arthralgia, myalgia, and fever with 76.7% rate of AEs
Machado *et al.*, 2018 (42)	19-53	*L. braziliensis*	IV meglumine antimoniate (15) IV meglumine antimoniate + topical tamoxifen (11) IV meglumine antimoniate + topical tamoxifen (12)	40% 36.4% 58%	24	Mild AEs were observed in all groups, with arthralgia, myalgia, and fever being the most common. One patient in the oral tamoxifen group dropped out due to headache and palpitations.
Machado *et al.*, 2021 (43)	33 36 33	*L. braziliensis*	IV meglumine antimoniate (45) Oral miltefosine (47) Oral miltefosine + topical GM-CSF (41)	44.4% 76.6% 75.6%	24	65% rate of AEs, such as arthralgia and/or myalgia, fever, and headache 76% rate of AEs, such as nausea, vomiting, and headache 79% rate of AEs, such as nausea, vomiting, and headache
Martinez *et al.*, 1992 (44)	11-40	*L. panamensis*	IV meglumine antimoniate (33) Oral allopurinol (25) Oral allopurinol + IV meglumine antimoniate (35)	36% 80% 74%	52	No major AEs

Martinez *et al.*, 1997 (45)	18-57	*L. braziliensis* and *L. panamensis*	IV sodium stibogluconate + oral allopurinol (51) IV sodium stibogluconate (49)	71% 39%	52	More cases of eosinophilia and rash, but no significant toxic effects One patient stopped treatment due to severe chemical hepatitis with neurological manifestations
Mendes *et al.*, 2021 (46)	18-65	*L. guyanensis*	Oral miltefosine (50) Oral miltefosine + topical GM-CSF (50) IV meglumine antimoniate (50)	66% 58% 50%	24	Common AEs were nausea (85%) and vomiting (39%) Frequent arthralgia (51%) and myalgia (48%). One severe AE led to treatment discontinuation
Miranda-Verástegui *et al.*, 2005 (47)	14.1 19.25	*L. peruviana* and *L. braziliensis*	IM or IV meglumine antimoniate + imiquimod cream (18) IM or IV meglumine antimoniate (20)	72% 75%	52	Neurotoxic effects of antimony therapy have been reported Mild erythema was more common, but no severe AEs were reported
Miranda-Verástegui *et al.*, 2009 (48)	25 25.9	*L. braziliensis*, *L. peruviana*, and *L. guyanensis*	IV sodium stibogluconate + imiquimod cream (40) IV sodium stibogluconate (40)	75% 58%	52	Similar mild AEs were noted in both groups.
Mohebali *et al.*, 2007 (49)	20.2 16.8	*L. major*	Oral miltefosine (28) IM meglumine antimoniate (30)	92.9% 83.3%	24	Nausea and vomiting were observed in miltefosine group. No significant difference in other AEs reporting was seen between the two groups.
Momeni *et al.*, 2002 (50)	20.2 23.6	*L. major*	IV or IM meglumine antimoniate (35) IV or IM meglumine antimoniate (low-dose) + oral allopurinol (31)	72.4% 80.6%	4	Skin eruption in one patient and generalized muscle pain and weakness in four patients Mild abdominal pain and nausea in six patients
Munir *et al.*, 2008 (51)	14-55	*L. tropica*	IM meglumine antimoniate (20) IM + IL meglumine antimoniate (20)	55% 75%	12	Combination group experienced more AEs, including myalgia, arthralgia, headache, urticaria, and one case of cardiomyopathy, which resolved after treatment discontinuation
Nassiri-Kashani *et al.*, 2005 (52)	12-60	*L. major*	Oral itraconazole (70)	95.7%	12	Gastrointestinal complaints and headaches

Navin *et al.*, 1992 (53)	20	*L. braziliensis* and *L. mexicana*	IV sodium stibogluconate (32)	90.6%	52	Elevated liver enzymes and electrocardiographic changes, but these normalized by week 9
			Oral ketoconazole (32)	56.3%		Gastrointestinal complaints
Neves *et al.*, 2011 (54)	29.3	*L. guyanensis* and *L. braziliensis*	IM pentamidine (58)	62.1%	24	35.1% experienced injection site pain
	28.5		IV or IM meglumine antimoniate (58)	60.3%		20.3% experienced arthralgia, 9.5% had headaches
Ozgoztasi *et al.*, 1997 (55)	≤ 10	Not specified	Oral ketoconazole (32)	0%	4	No AEs
Parhizkar *et al.*, 2024 (56)	29.8	*L. major*	Oral fluconazole + cryotherapy (28, 49 lesions)	53.1%	4	No AEs
Pennini *et al.*, 2023 (57)	37.9	*L. guyanensis*	IM pentamidine (24)	100%	24	Mild AEs related to pentamidine were reported, such as bitter taste, dyspepsia, abdominal pain, and mild hypoglycemia. There were no AEs related to tamoxifen.
	36.6		IM pentamidine + oral tamoxifen (25)	72%		
Prates *et al.*, 2017 (58)	27.9	*L. braziliensis*	Oral fluconazole (27)	22.2%	24	Mild AEs with similar frequencies in both groups (fluconazole: 37%, sodium stibogluconate: 34.6%). One patient in the fluconazole group discontinued due to malaise, headache, and dizziness.
	35		IV sodium stibogluconate (26)	53.8%		
Rahman *et al.*, 2023 (59)	28.82	Not specified	IM meglumine antimoniate (33)	33.3%	4	Injection site pain (39.3%), abscess formation (21.2%), and ECG changes (24.2%)
	32.09		Oral miltefosine (33)	12.1%		Nausea and vomiting (63.6%), abdominal fullness (33.3%), and fever (15.2%)
Ribeiro *et al.*, 2021 (60)	18-65	*L. braziliensis*	IV meglumine antimoniate (14)	43%	24	2 patients with discontinuation of treatment protocol due to AEs: one from the oral tamoxifen group and the other from the topical tamoxifen group
			IV meglumine antimoniate + oral tamoxifen (10)	70%		
			IV meglumine antimoniate + topical tamoxifen (10)	40%		
Rubiano *et al*, 2012 (61)	2-12	*L. panamensis* and *L. guyanensis*	Oral miltefosine (55)	87.3%	26	Nausea, vomiting, and abdominal pain
			IM meglumine antimoniate (56)	71.4%		Elevated liver enzymes, fever, and loss of appetite

Sadeghian *et al.*, 2006 (62)	27	*L. major*	IM meglumine antimoniate + oral pentoxifylline (32)	81.3%	12	No significant AEs
	31		IM meglumine antimoniate (31)	51.6%		A pruritic rash in one patient
Saenz *et al.*, 1990 (63)	25	*L. panamensis*	Oral ketoconazole (22)	73%	52	Mild, reversible hepatocellular enzyme elevation in 27% of patients, and a reversible decrease in serum testosterone levels
	34		IM sodium stibogluconate (19)	68%		47% had mild hepatocellular enzyme elevation, and most patients experienced pain at the injection site, myalgia, or headache
Salmanpour *et al.*, 2001 (64)	20.7	*L. tropica* and *L. major*	Oral ketoconazole (64)	89%	24	Mild nausea and abdominal pain in some patients; two cases of increased liver enzymes (resolved post-treatment)
Sattar *et al.*, 2020 (65)	26.76	Not specified	Oral zinc sulphate (34)	58.8%	6	Mild nausea and abdominal pain in some patients
Shahzad *et al.*, 2020 (66)	34.26	Not specified	Oral azithromycin (28)	67.9%	12	No significant AEs
Shanehsaz *et al.*, 2015 (67)	25.2	Not specified	IM meglumine antimoniate (30, 90 lesions)	91.1%	12	Higher incidence of AEs, including musculoskeletal pain, nausea, and minor skin reactions
			IM meglumine antimoniate (low-dose) + oral cimetidine (30, 150 lesions)	84.7%		Slightly higher AEs rate compared to low-dose meglumine antimoniate alone
			IM meglumine antimoniate (low-dose) (30, 120 lesions)	78.3%		Lower incidence of AEs
Soto *et al.*, 2004 (68)	24.9	*L. panamensis* and *L. braziliensis* and *L. mexicana*	Oral miltefosine (82)	73.2%	24	Nausea (36%) and vomiting (31%), with mild increases in creatinine observed in 33% of participants
Soto-Mancipe *et al.*, 1993 (69)	18-25	*L. panamensis* and *L. braziliensis* and *L. mexicana*	IM meglumine antimoniate (23)	91%	52	Nine patients (39%) reported subjective. AEs, including arthralgia and myalgia.
			IM pentamidine (24)	96%		Eight patients (30%) reported mild to moderate subjective AEs, including myalgias, nausea and headache.
			Oral itraconazole (20)	5%		Not reported.

Toledo Junior *et al.*, 2014 (70)	14-65	*L. braziliensis*	IM or IV meglumine antimoniate (18) Oral azithromycin (21)	72.2% 23.8%	12	Higher AE incidence (62.5%), including myalgia and arthralgia Fewer AEs, mainly gastrointestinal issues
Vélez *et al.*, 2010 (71)	19-38	*L. panamensis* and *L. braziliensis*	Oral miltefosine (122) IM meglumine antimoniate (121)	69.8% 85.1%	24	Gastrointestinal symptoms such as nausea, vomiting, and abdominal pain Musculoskeletal pain, fever, and higher organ toxicity
Yazdanpanah *et al.*, 2011 (72)	32 25	*L. tropica*	Oral zinc sulfate (26, 43 lesions) IM meglumine antimoniate (74, 127 lesions)	30.2% 35.5%	6	No significant AEs Serious AEs, such as severe muscle pain and injection site reactions, leading to discontinuation for some patients
Zabolinejad *et al.*, 2022 (73)	31.8 35.8	Not specified	Oral clarithromycin (10, 49 lesions) IM or IV meglumine antimoniate (10, 45 lesions)	100% 96.3%	52	No AEs reported in either group
Zerehsaz *et al.*, 1999 (74)	1.5-69	Not specified	IM meglumine antimoniate (85)	27.1%	6	Cutaneous AEs, including urticaria and generalized pruritus

AEs, Adverse events; IM, intramuscular; IL, intralesional; IV, intravenous; GM-CSF, granulocyte-macrophage colony-stimulating factor; ECG, electrocardiogram.

Table S2. Risk of Bias assessment of randomized clinical trials. The risk-of-bias tool version 2 (RoB2), which is the recommended tool to assess the risk of bias in randomized trials included in Cochrane Reviews, was applied for the evaluation. D1, Randomization process; D2, deviations from the intended interventions; D3, missing outcome data; D4, measurement of the outcome; D5, selection of the reported result; Green, low risk; Yellow, some concerns; Red, high risk.

Study	Risk of bias					
	D1	D2	D3	D4	D5	Overall
Abtahi-Naeini *et al.*, 2021 (1)	!	+	+	+	!	!
Adam *et al.*, 2009 (2)	+	+	+	+	!	!
Ahmed *et al.*, 2022 (3)	+	+	+	+	!	!
Alfahaad, 2020 (4)	!	+	-	!	!	-
Al-Fouzan *et al.*, 1991 (5)	!	+	+	+	!	!
Alkhawajah *et al.*, 1997 (6)	!	!	-	+	!	-
Almeida *et al.*, 1999 (7)	!	+	+	+	!	!
Alrajhi *et al.*, 2002 (8)	+	+	+	+	!	!
Arana *et al.*, 1994 (9)	!	+	+	+	!	!
Arevalo *et al.*, 2007 (10)	!	+	+	!	!	!
Armijos *et al.*, 2004 (11)	+	+	!	+	!	!
Aronson *et al.*, 2010 (12)	+	+	+	+	+	+
Bashir *et al.*, 2019 (13)	+	+	!	+	+	!
Brito *et al.*, 2014 (14)	!	+	+	+	!	!
Brito *et al.*, 2017 (15)	!	+	+	+	!	!
Castro *et al.*, 2022 (16)	+	+	+	+	+	+
Celes *et al.*, 2022 (17)	+	+	+	+	+	+
Chrusciak-Talhari *et al.*, 2011 (18)	+	+	+	+	+	+
D'Oliveira Jr. *et al.*, 1997 (19)	!	+	-	+	!	-
Dastgheib *et al.*, 2012 (20)	!	+	+	-	!	-
Dogra *et al.*, 1990 (21)	!	+	+	-	!	-
Dogra *et al.*, 1994 (22)	+	+	+	+	!	!
Dogra *et al.*, 1996 (23)	!	-	+	!	!	-
El-Sayed *et al.*, 2010 (24)	!	+	+	!	+	!
Esfandiarpour *et al.*, 2002 (25)	!	+	+	!	!	!
Farajzadeh *et al.*, 2015 (26)	+	+	+	+	+	+
Firooz *et al.*, 2006 (27)	+	+	+	+	+	+
Hanif *et al.*, 2016 (28)	+	+	+	!	+	!
Hepburn *et al.*, 1994 (29)	!	+	+	!	+	!
Jaffar *et al.*, 2006 (30)	!	+	+	+	+	!
Khattak *et al.*, 2021 (31)	!	+	+	!	+	!
Kochar *et al.*, 2000 (32)	+	+	+	+	+	+
Kochar *et al.*, 2006 (33)	!	+	+	+	+	!
Krolewiecki *et al.*, 2007 (34)	+	+	+	!	!	!
Lago *et al.*, 2024 (35)	+	+	+	!	+	!
Layegh *et al.*, 2007 (36)	!	+	+	!	+	!
Lobo *et al.*, 2006 (37)	!	+	+	!	!	!
López *et al.*, 2012 (38)	+	+	+	!	+	!
López *et al.*, 2022 (39)	+	+	+	+	+	+
Lyra *et al.*, 2023 (40)	+	+	+	!	+	!
Machado *et al.*, 2010 (41)	+	+	+	+	+	+
Machado *et al.*, 2018 (42)	+	+	+	+	+	+
Machado *et al.*, 2021 (43)	+	+	+	+	+	+
Martinez *et al.*, 1992 (44)	+	+	+	!	+	!
Martinez *et al.*, 1997 (45)	+	+	+	!	+	!
Mendes *et al.*, 2021 (46)	+	+	+	!	+	!

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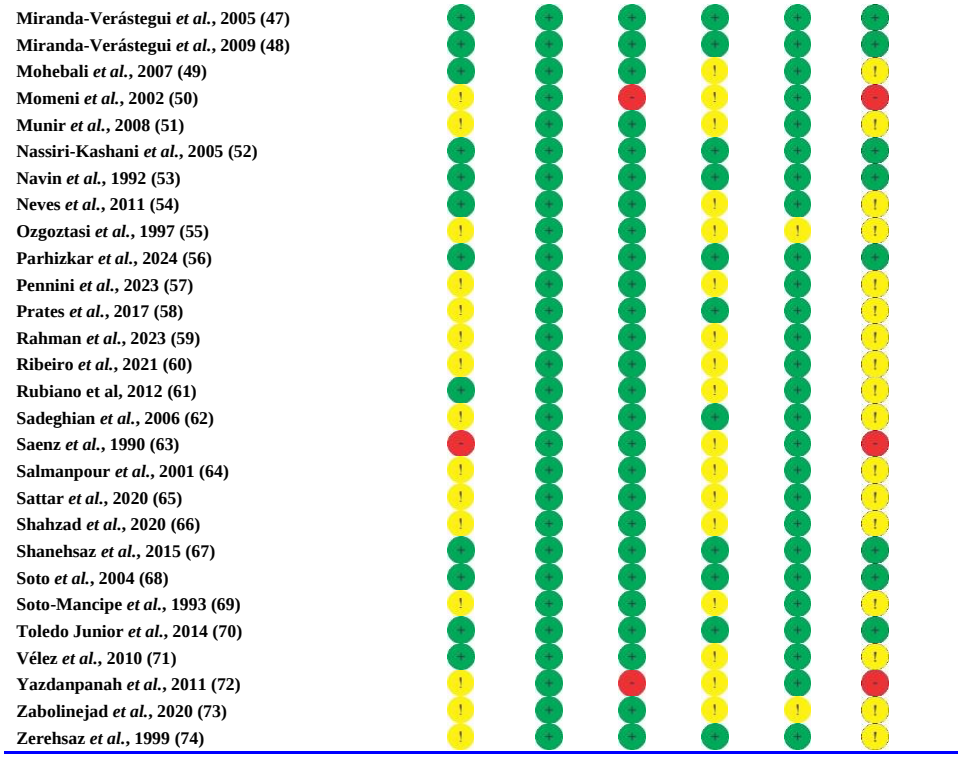

Miranda-Verástegui et al., 2005 (47)
Miranda-Verástegui et al., 2009 (48)
Mohebali et al., 2007 (49)
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Munir et al., 2008 (51)
Nassiri-Kashani et al., 2005 (52)
Navin et al., 1992 (53)
Neves et al., 2011 (54)
Ozgoztasi et al., 1997 (55)
Parhizkar et al., 2024 (56)
Pennini et al., 2023 (57)
Prates et al., 2017 (58)
Rahman et al., 2023 (59)
Ribeiro et al., 2021 (60)
Rubiano et al, 2012 (61)
Sadeghian et al., 2006 (62)
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Sattar et al., 2020 (65)
Shahzad et al., 2020 (66)
Shanehsaz et al., 2015 (67)
Soto et al., 2004 (68)
Soto-Mancipe et al., 1993 (69)
Toledo Junior et al., 2014 (70)
Vélez et al., 2010 (71)
Yazdanpanah et al., 2011 (72)
Zabolinejad et al., 2020 (73)
Zerehsaz et al., 1999 (74)

Table S3. Node-splitting analysis for local inconsistency under the random-effects model for the sensitivity NMA (excluding high risk studies)

Treatment comparison	k	Direct evidence proportion	NMA RR	Direct RR	Indirect RR	RoR	z	*P*-value
Systemic allopurinol versus placebo/control	1	0.07	4.77	28.38	4.22	6.73	1.29	0.197
Systemic azithromycin versus placebo/control	1	0.53	1.61	1.58	1.63	0.97	-0.05	0.958
Systemic fluconazole versus placebo/control	1	0.76	1.87	2.33	0.94	2.48	1.26	0.207
Systemic ketoconazole versus placebo/control	2	0.82	1.9	1.88	1.98	0.95	-0.06	0.954
Systemic meglumine antimoniate versus placebo/control	3	0.24	3.56	6.95	2.88	2.41	1.55	0.121
Systemic meglumine antimoniate + systemic allopurinol versus	1	0.05	5.61	26.5	5.17	5.13	1.11	0.265
Systemic miltefosine versus placebo/control	2	0.55	3.88	2.45	6.72	0.36	-2.09	**0.037**
Systemic pentamidine versus placebo/control	1	0.19	3.89	7.67	3.32	2.31	1.04	0.3
Systemic sodium stibogluconate versus placebo/control	2	0.63	2.92	1.87	6.23	0.3	-1.8	0.072
Systemic allopurinol versus systemic meglumine antimoniate	2	0.82	1.34	1.45	0.94	1.55	0.61	0.54
Systemic allopurinol versus systemic meglumine antimoniate + systemic allopurinol	2	0.92	0.85	0.76	3.45	0.22	-1.59	0.112
Systemic azithromycin versus systemic meglumine antimoniate	2	0.7	0.45	0.46	0.44	1.03	0.05	0.958
Systemic fluconazole versus systemic sodium stibogluconate	1	0.51	0.64	0.41	1.02	0.4	-1.26	0.207
Systemic itraconazole versus systemic meglumine antimoniate	1	0.12	0.33	0.06	0.41	0.16	-1.71	0.088
Systemic itraconazole versus systemic pentamidine	1	0.16	0.3	0.06	0.41	0.15	-1.7	0.088
Systemic ketoconazole versus systemic sodium stibogluconate	2	0.94	0.65	0.74	0.09	7.82	1.8	0.071
Systemic meglumine antimoniate versus systemic meglumine antimoniate + systemic allopurinol	3	0.99	0.63	0.65	0.09	7.48	0.88	0.378
Systemic meglumine antimoniate versus systemic miltefosine	8	0.93	0.92	0.86	2.36	0.36	-2.09	**0.037**
Systemic meglumine antimoniate versus systemic pentamidine	2	0.97	0.92	0.99	0.08	11.68	1.9	0.057

NMA, Network meta-analysis; RR, relative risk; RoR, ratio of ratios; k, number of studies included in the comparison; z, z-statistic; *P*-value, probability value.

Table S4. Node-splitting analysis for local inconsistency under the random-effects model for the overall NMA

Treatment comparison	k	Direct evidence proportion	NMA RR	Direct RR	Indirect RR	RoR	z	*P*-value
Systemic allopurinol versus placebo/control	1	0.06	4.08	28.38	3.58	7.93	1.4	0.162
Systemic azithromycin versus placebo/control	1	0.53	1.63	1.58	1.67	0.95	-0.1	0.923
Systemic fluconazole versus placebo/control	1	0.75	1.87	2.33	0.97	2.39	1.21	0.226
Systemic ketoconazole versus placebo/control	3	0.79	2.29	2.18	2.73	0.8	-0.29	0.772
Systemic meglumine antimoniate versus placebo/control	3	0.24	3.63	6.95	2.95	2.35	1.5	0.134
Systemic meglumine antimoniate + systemic allopurinol versus placebo/control	1	0.04	5	26.5	4.62	5.73	1.19	0.233
Systemic miltefosine versus placebo/control	2	0.54	3.94	2.45	6.95	0.35	-2.13	**0.033**
Systemic pentamidine versus placebo/control	1	0.19	4	7.67	3.42	2.24	0.99	0.322
Systemic sodium stibogluconate versus placebo/control	3	0.63	2.94	2.23	4.68	0.48	-1.15	0.249
Systemic allopurinol versus systemic meglumine antimoniate	3	0.78	1.12	1.25	0.77	1.63	0.76	0.45
Systemic allopurinol versus systemic meglumine antimoniate + systemic allopurinol	2	0.88	0.82	0.75	1.46	0.51	-0.85	0.393
Systemic azithromycin versus systemic meglumine antimoniate	2	0.7	0.45	0.45	0.43	1.06	0.1	0.923
Systemic fluconazole versus systemic sodium stibogluconate	1	0.5	0.64	0.41	0.99	0.42	-1.21	0.226
Systemic itraconazole versus systemic meglumine antimoniate	1	0.12	0.39	0.06	0.5	0.13	-1.89	0.059
Systemic itraconazole versus systemic pentamidine	1	0.16	0.36	0.06	0.5	0.12	-1.89	0.059
Systemic ketoconazole versus systemic sodium stibogluconate	3	0.96	0.78	0.84	0.1	8.28	1.83	0.067
Systemic meglumine antimoniate versus systemic meglumine antimoniate + systemic allopurinol	4	0.99	0.73	0.72	2.77	0.26	-0.82	0.415
Systemic meglumine antimoniate versus systemic miltefosine	8	0.93	0.92	0.86	2.44	0.35	-2.13	**0.033**
Systemic meglumine antimoniate versus systemic pentamidine	2	0.97	0.91	0.99	0.08	13.15	1.98	**0.048**

NMA, Network meta-analysis; RR, relative risk; RoR, ratio of ratios; k, number of studies included in the comparison; z, z-statistic; *P*-value, probability value.

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